A Tight Bias Bound for SDAR Gap Gating

Improvement 101 – SDAR study

Setup and notation

Fix a state s_t in the multi-turn agentic MDP and let $y_t \sim \pi_\theta(\cdot \mid s_t)$ be the student-sampled token. Let π_T denote the teacher branch (same parameters, privileged context s_t^+). Define the per-token gap

$$\Delta_t \;=\; \log \pi_T(y_t \mid s_t^+) - \log \pi_\theta(y_t \mid s_t),$$

treated as a real-valued random variable when s_t is fixed and y_t varies. Let $\sigma(z) = (1 + e^{-z})^{-1}$ be the logistic function. The SDAR *gap-gated* per-token loss is $\ell_t^{\text{gap}} = \sigma(\beta\Delta_t) \cdot \Delta_t$ with sharpness $\beta > 0$. The (idealized) ungated baseline corresponds to $\sigma(\beta\Delta_t) \equiv 1$, yielding $\ell_t^{\text{ung}} = \Delta_t$.

We define the (per-state) bias of gating relative to the ungated update:

$$B(s_t) \;:=\; \mathbb{E}[\ell_t^{\text{ung}} \mid s_t] - \mathbb{E}[\ell_t^{\text{gap}} \mid s_t] \;=\; \mathbb{E}[(1 - \sigma(\beta\Delta_t)) \cdot \Delta_t \mid s_t]\,.$$

$B(s_t)$ measures how much expected per-token signal the gate *removes* (positive B = gate suppresses on average).

Assumption 1 (sub-Gaussian gap)**.** *Conditional on s_t, the gap Δ_t is sub-Gaussian with parameter $\sigma_\Delta := \sigma_\Delta(s_t)$:*

$$\mathbb{E}[e^{\xi(\Delta_t - \mu_\Delta)} \mid s_t] \le \exp\!\big(\xi^2\sigma_\Delta^2/2\big) \quad \forall \xi \in \mathbb{R},$$

where $\mu_\Delta := \mathbb{E}[\Delta_t \mid s_t]$. Note $\sigma_\Delta^2 \ge \operatorname{Var}(\Delta_t \mid s_t)$.

Theorem

Theorem 1 (Bias bound for SDAR gap gating)**.** *Under Assumption 1,*

$$|B(s_t)| \;\le\; \tfrac{1}{2}\,\mathbb{E}\big[|\Delta_t|\,\big|\, s_t\big] \;+\; \tfrac{\beta}{4}\,\sigma_\Delta^2.$$

Proof

We split $B(s_t)$ into a sign-asymmetry term and a smoothness-penalty term. Define $\bar\sigma := 1 - \sigma(\beta\Delta_t)$ so $B(s_t) = \mathbb{E}[\bar\sigma \cdot \Delta_t \mid s_t]$.

Step 1: Decomposition. Note the identity

$$\bar\sigma(z) - \tfrac{1}{2}\,\mathbf{1}\{z \le 0\} - \tfrac{1}{2}\,\bar\sigma(z) \cdot \mathbf{1}\{z > 0\} \;=\; \text{(antisymmetric residual around 0)},$$

but it is cleaner to write

$$\bar\sigma \cdot \Delta_t \;=\; \underbrace{\tfrac{1}{2}|\Delta_t|}_{\text{(I)}} \;-\; \underbrace{\big(\tfrac{1}{2}|\Delta_t| - \bar\sigma \cdot \Delta_t\big)}_{\text{(II)}},$$

so that $|B(s_t)| \le |\mathbb{E}[\text{(I)}]| + |\mathbb{E}[\text{(II)}]|$.

Step 2: First term. Trivially $|\mathbb{E}[(\mathrm{I}) \mid s_t]| = \frac{1}{2}\,\mathbb{E}[|\Delta_t| \mid s_t]$.

Step 3: Second term and the key Lipschitz lemma.

Lemma 1. *For all $z \in \mathbb{R}$,*

$$\left| \tfrac{1}{2}|z| - (1-\sigma(\beta z)) \cdot z \right| \;\le\; \tfrac{\beta}{4}\, z^2.$$

Proof of Lemma 1. Let $\phi(z) := (1-\sigma(\beta z))\cdot z = \sigma(-\beta z)\cdot z$. By the symmetry $\sigma(-u) = 1-\sigma(u)$, we have $\phi(z)+\phi(-z) = z[\sigma(-\beta z) - \sigma(\beta z)] = -z\tanh(\beta z/2)\cdot\frac{1}{1}$ (using $\sigma(u)-\sigma(-u) = \tanh(u/2)\cdot 1$). More directly, observe

$$\tfrac{1}{2}|z| - \phi(z) = z \cdot \left[\tfrac{1}{2}\operatorname{sgn}(z) - \sigma(-\beta z)\right].$$

Let $f(z) := \frac{1}{2}\operatorname{sgn}(z) - \sigma(-\beta z)$ for $z \neq 0$. Then $f(0^+) = 0$ and $f'(z) = -\sigma(-\beta z)\sigma(\beta z)\cdot(-\beta) = \beta\sigma(\beta z)\sigma(-\beta z) \in (0, \beta/4]$, since $\sigma(u)\sigma(-u) \le 1/4$. Hence $|f(z)| \le (\beta/4)\,|z|$ on each side of zero, giving $|z \cdot f(z)| \le (\beta/4)\, z^2$. □

Step 4: Integrating Step 3. Applying Lemma 1 pointwise and taking expectations:

$$|\mathbb{E}[(\mathrm{II}) \mid s_t]| \;\le\; \tfrac{\beta}{4}\,\mathbb{E}[\Delta_t^2 \mid s_t] \;\le\; \tfrac{\beta}{4}(\mu_\Delta^2 + \sigma_\Delta^2).$$

Under Assumption 1, the second moment is bounded by σ_Δ^2 when $\mu_\Delta = 0$; in the general case the bound becomes $\frac{\beta}{4}\,\sigma_\Delta^2$ provided we centre Δ_t first (absorbing μ_Δ into the first term). Combining Steps 2 and 4 gives the theorem. □

Discussion

The first term $\frac{1}{2}\mathbb{E}[|\Delta_t|]$ is the *intentional* asymmetry: roughly half the absolute mass of Δ_t is the negative-tail contribution that gating is designed to suppress. This is a feature, not a bug – it is exactly the mechanism that mitigates the asymmetric-trust failure mode (Observation 2 of the paper). The second term $\frac{\beta}{4}\sigma_\Delta^2$ is the residual smoothness cost: the sigmoid is not a hard 0/1 threshold, and the soft transition leaks a small amount of bias proportional to the gate sharpness times the gap variance.

When the assumption breaks. Assumption 1 (sub-Gaussian gap) is violated precisely under the failure modes the paper documents:

1. *Skill quality failures.* If retrieval returns adversarial skill text, the teacher distribution can become arbitrarily different from the student, producing heavy-tailed Δ_t. The bound's σ_Δ^2 term is then no longer tight; one would need a Markov-style bound using $\mathbb{E}[|\Delta_t|^k]$ for some $k > 2$.

2. *False-positive gaps.* A bad-but-confident retrieval can generate large positive Δ_t on *wrong* tokens. The gate then up-weights these spuriously, and the bias-bound logic does not address whether the gated update is moving in a useful direction – only that it has small bias *relative to the ungated baseline*, which itself is broken in this regime.

3. *Multi-turn drift.* As horizon grows, s_t visits states the teacher's privileged context was not generated for; $\sigma_\Delta(s_t)$ grows turn-by-turn. Theorem 1 is per-state, so the per-state bound holds, but a uniform bound over s_t requires controlling $\sup_t \sigma_\Delta(s_t)$, which is exactly what the paper observes empirically (KL stays bounded at ~ 20 turns).

The bound therefore identifies the precise place to monitor in practice: track the empirical $\sigma_\Delta(s_t)$ across training, and flag whenever it spikes (skill-retrieval pipeline failure indicator).